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Purification and Some Properties of Tobacco Streak and Tulare Apple Mosaic Viruses

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Tobacco streak virus (TSV) was stabilized in extracts by 2-mercaptoethanol and, after clarification of the extracts with hydrated calcium phosphate, concentrated by high speed centrifugation. Host protein was removed from concentrated preparations by acid precipitation and by precipitation with γ-globulin of antiserum to host protein. Purification was completed by two cycles of high (40,000 rpm) and low (10,000) speed centrifugation. TSV measured about 28 mμ in diameter. Centrifugation of purified virus on sucrose density gradients resulted in two closely spaced zones, only the lower of which was infective. Material from the upper zone enhanced infectivity of infective virus; enhancement was more pronounced on the host on which the infectivity-dilution curve was abnormally steep. Tulare apple mosaic virus was purified by the procedure used for TSV. It was slightly larger (32–34 mμ in diameter) than TSV, reacted with homologous antiserum but not with TSV antiserum, and was separable on sucrose density gradients into infectious and noninfectious zones.

INTRODUCTION

There were several reasons for developing a purification method for tobacco streak virus (TSV). The virus is unstable in crude extracts and apparently is permanently inactivated by oxidized phenolic compounds (Fulton, 1949) in tissue extracts. Inactivation can be prevented by antioxidants, which should make it possible to retain infectivity while separating the virus from deleterious tissue components. Purified virus, if relatively stable, would provide favorable material for investigating the nature of the inactivation in tissue extracts.

Another reason for attempting purification of TSV was that in preliminary trials of several plant species as local lesion assay hosts, the dilution infectivity curves obtained with some hosts were abnormally steep and resembled those reported for two *Prunus* viruses (Fulton, 1962). TSV offers some advantages over the stone fruit viruses for investigating the nature of such multiple-hit dilution curves. The infectivity of extracts of TSV-infected tobacco tissue suggested that they contained more virus than could be obtained in tissue infected with stone fruit viruses. TSV also exists in a number of strains that produce distinctive symptoms on tobacco and other hosts.

TSV has been recorded from a wide variety of plants and from widely scattered locations (Costa and Carvalho, 1961; Patino and Zaumeyer, 1959; Devergne, 1961; Thomas and Zaumeyer, 1950). Antiserum produced from purified virus would be useful for identifying suspected isolates. Scott *et al.* (1961) prepared an antiserum to the bean red node strain of TSV with virus partially purified by chloroform-butanol extraction and alternating high and low speed centrifugation. The antiserum, however, had a fairly high titer of antibody to host protein.

Mink *et al.* (1966) purified the bean red node strain of TSV from bean stem tissue by means of a charcoal-freezing process. The method, however, was reported as leaving pigment in preparations from leaf tissue.

After an effective purification method had

been developed for TSV, it was applied to Tulare apple mosaic virus (TAMV), which has many properties similar to TSV (Mink *et al.*, 1963).

MATERIALS AND METHODS

Three isolates of TSV originally were found in tobacco. These ranged in severity on tobacco from typical (strain HF), to medium (McC), to mild (M). One isolate (WC) was obtained from white clover. Another isolate (DV) was one of several obtained from naturally infected *Dahlia variabilis* (Willd.) Desf. and reproduced the original chlorotic bands and rings when inoculated to seedlings of this plant. The bean red node strain was obtained from W. J. Zaumeyer. Tulare apple mosaic virus was obtained from C. E. Yarwood. All isolates were kept in diced tobacco tissue that had been dehydrated cold in a vacuum. Some such preparations of TSV have remained infective for over 18 years at 2–4°.

Inoculum was produced in *Nicotiana tabacum* L. cv. Xanthi-nc. For purification, leaves of *N. tabacum* cv. T.I.787 were dusted with Corundum, inoculated heavily, and harvested 4–6 days later, which was 2–3 days after local symptoms had appeared. Inoculum was prepared by grinding infected leaves in 0.03 *M* phosphate buffer, pH 8.0, containing 0.02 *M* 2-mercaptoethanol as an antioxidant.

All plants were grown in a greenhouse at 24–25°. During the winter and whenever light intensity was low, inoculated leaves often became severely necrotic. Necrosis could be reduced by illuminating inoculated plants for 16–18 hours a day. Most virus for purification was produced at this day-length in temperature control chambers with fluorescent lighting.

Hosts used for assaying infectivity of TSV were *Dolichos biflorus* L., *D. lablab* L., and *Vigna cylindrica* (L.) Skeels. TAMV was assayed on *Phaseolus vulgaris* cv. Bountiful. In assaying infectivity, replicates of three or four treatments were randomized among the 12 halves of six primary leaves of three plants for each trial.

RESULTS

Several antioxidants were compared for their ability to stabilize TSV infectivity in the fairly concentrated plant extracts that would be necessary for purification. The most effective was 2-mercaptoethanol (2-ME) (Table 1). Sodium diethyldithiocarbamate, also an inhibitor of polyphenol oxidase, had no stabilizing effect.

Clarification of Infectious Extracts

Attempts were made to clarify tobacco leaf extracts with butanol, but as little as 4% caused appreciable loss of infectivity. The virus was not adsorbed, however, on hydrated calcium phosphate (HCP) (Fulton, 1959), and this material was used for initial clarification. After midribs were removed, infected leaves were chilled and homogenized in cold phosphate buffer in a Waring Blendor for 5 minutes. Antifoam A (Dow-Corning) was added to prevent foaming. The proportion (by weight) of leaves to buffer, and the concentration of buffer affected the degree of clarification obtained. A volume of 0.02 *M*, pH 8.0 buffer equivalent to 1.5 times the weight of tissue was most satisfactory.

To stabilize infectivity, 2-ME was added to the buffer, before the leaves were homogenized, to make an 0.02 *M* solution. Although sodium diethyldithiocarbamate did

TABLE 1

STABILITY OF TOBACCO STREAK VIRUS IN 0.03 *M*, pH 8.0 PHOSPHATE BUFFER WITH ADDED CHEMICALS AT 0.02 *M*[a]

Added chemical	Lesions/half-leaf, average of 3 or more trials	
	Inoculated immediately	Inoculated after 4 hours
Ascorbic acid	40	19
Cysteine-HCl	42	30
Hydroxylamine-HCl	50	12
2-Mercaptoethanol	54	52
Sodium diethyldithiocarbamate	33	1
Sodium hydrosulfite	16	0
Sodium sulfite	35	4
Sodium thioglycolate	47	0

[a] Six 8-mm leaf disks were ground in 1 ml, then diluted 1:10 in buffer immediately before inoculation.

not stabilize infectivity, it was also added to the buffer to make 0.02 *M*. Without it the virus pellets were brown.

The leaf homogenate was centrifuged in 250-ml bottles for 20 minutes at 3000 rpm in an International model UV centrifuge. The supernatant liquid was then stirred 3 minutes with a volume of HCP equivalent to 0.8 of the original weight of tissue. When tissue was rather necrotic, or when an 0.8 portion failed to clarify the extract completely, an additional portion (0.1 of the tissue weight) was used. This mixture was centrifuged 20 minutes at 3000 rpm.

The clear yellow supernatant liquid from the HCP clarification was centrifuged 2.5 hours at 30,000 rpm in a Spinco No. 30 rotor. The resulting pellets were resuspended in 0.02 *M* phosphate buffer, pH 8.0, by shaking on a mechanical shaker at 2°. The volume for resuspension was as small as practical, usually 0.5 ml per tube.

The resuspended pellets were centrifuged for 20 minutes at 10,000 rpm in a Spinco No. 40 rotor. This removed some of the insoluble material. The supernatant liquid contained the virus and appreciable amounts of host protein, not all of which was removed by subsequent cycles of high and low speed centrifugation. Other methods were tried, therefore, in attempts to remove this extraneous material.

Acidification of the resuspended pellets to pH 5.2 with HCl, H_3PO_4, or citric acid was fairly effective. The virus withstood the acidity for the time required to remove the precipitate by centrifugation. Contaminating material was not completely removed, but acidification with citric acid was used with many preparations to simplify subsequent purification steps.

Heating the resuspended pellets to 38° for 75 minutes or adding 165 mg $(NH_4)SO_4$ per milliliter also precipitated extraneous material and left the virus in the supernatant liquid. Neither treatment, however, removed all the contaminating protein. Magnesium-saturated bentonite (Dunn and Hitchborn, 1965), with several concentrations of magnesium ion, adsorbed the virus.

Nonvirus material was most efficiently removed by precipitating it with the γ-globulin portion of an antiserum prepared with tobacco protein. The procedure was modified slightly from that described by Gold (1961).

Preparation of Antiserum to Tobacco Protein

It seemed probable that the contaminating host protein consisted of particulate material which sedimented during high speed centrifugation. Antigen was prepared, therefore, by clarifying healthy tobacco leaf tissue with HCP in the same manner as virus preparations were clarified. The yellowish supernatant liquid was centrifuged for 5–6 hours at 30,000 rpm in a Spinco No. 30 rotor. The resulting small clear pellets readily resuspended in phosphate buffer and gave a typical protein UV absorption spectrum, with a peak at 280 mμ and an $A_{260:280}$ ratio of 0.70 or less.

Rabbits were injected intramuscularly every 3 days with 1 ml of the protein preparation (A_{260} ca. 50) emulsified in Freund's incomplete adjuvant. After two to four intramuscular injections, the same amount of protein, without adjuvant, was injected intravenously every 3 days. After eight to ten injections, antiserum diluted 1:1280 or more precipitated appropriately diluted tobacco protein.

Rabbits were bled from the ear three or four times, after antiserum titers were sufficiently high, with continued injections between bleedings. After a lapse of 2 or 3 months a second series of injections was given, with half the previous amount of protein in Freund's incomplete adjuvant for the first one or two injections.

The γ-globulin was precipitated from the tobacco antiserum with 40% saturated $(NH_4)SO_4$. The precipitate was resuspended and centrifuged twice in fresh $(NH_4)SO_4$ solution, then dissolved in 0.14 *M* NaCl equal to half the original volume and stored frozen in 1-ml lots.

In removing contaminating host protein from virus preparations, a portion of the γ-globulin (usually 1 ml) was added to the resuspended high speed pellets. Initial additions of antiserum resulted in a clouding. After 30 minutes' incubation, pellets obtained by centrifuging the sample for 20 minutes at 10,000 rpm were usually clear and glassy. Antiserum was added in incre-

ments until no precipitate formed. Adding the final increment, which removed all precipitable protein, caused a flocculant precipitate which settled on standing.

Virus was removed from the liquid, after host antigen was completely removed, by centrifugation at 40,000 rpm for 75 minutes in a Spinco No. 40 rotor. The total additions of γ-globulin represented 25–35% of the volume of the preparation at this stage. Virus centrifuged in the presence of this much serum protein formed pellets that could be resuspended only by prolonged shaking, so the virus serum mixture was diluted with 2–3 volumes of 0.02 *M* phosphate buffer, pH 8.0, before centrifugation.

Tests for Purity of Virus Preparations

Two cycles of high and low speed centrifugation removed all detectable rabbit protein. Virus preparations did not react with goat antirabbit serum. Antiserum to TSV was prepared by injecting intramuscularly 1 ml of virus (A_{260} ca. 5) emulsified in Freund's incomplete adjuvant. Seven or eight injections at 3-day intervals usually resulted in antiserum that reacted at a dilution of 1:1280. This antiserum did not react with healthy tobacco protein. In agar diffusion tests only one zone of precipitation appeared between virus and antiserum wells. Antiserum was prepared only to TSV strain HF. Each of the other TSV strains reacted with this antiserum, but TAMV did not, nor did TSV react with antiserum that precipitated TAMV.

The UV absorption spectrum of purified TSV was typical of nucleoprotein. The $A_{260:280}$ ratios averaged 1.48 for 15 preparations, with a high of 1.55 and a low of 1.41. Preparations with an A_{260} of 1.0 contained an average of 169 μg of virus per milliliter, determined by weighing dried material.

Infectivity of Purified Virus

Infectivity of purified preparations of TSV was usually assayed at four 2-fold dilutions on *Dolichos biflorus*. Preparations diluted to an A_{260} of 0.001–0.002 produced from one to five lesions per half-leaf. More concentrated preparations produced correspondingly more lesions; the dilution curve appeared normal. Lesion counts on *D. lablab* and *Vigna cylindrica*, however, resulted in unusually steep dilution curves (Fig. 1) resembling the curves described for cherry necrotic ringspot and prune dwarf viruses (Fulton, 1962). The difference in response between *D. biflorus* and *V. cylindrica* was particularly striking when the same dilutions were inoculated to both. On *V. cylindrica* the most infectious dilutions produced twice as many lesions as on *D. biflorus*, whereas the more dilute ones produced less than half as many lesions. The response of these species was the same when inoculated with TSV in crude tissue extracts in buffer containing 2-ME.

Stability of Purified Virus

In 0.02 *M*, pH 8.0 phosphate buffer, TSV retained much of the original infectivity for several months at 2°. In one experiment,

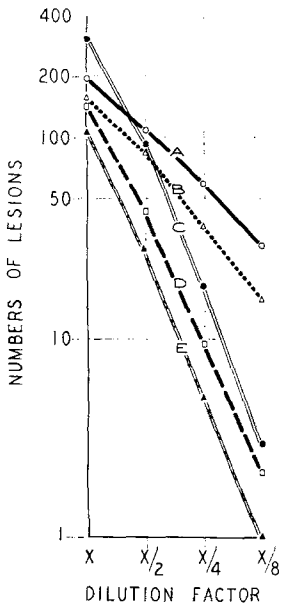


FIG. 1. Infectivity dilution curves produced by tobacco streak and Tulare apple mosaic viruses. Curve *A* represents the slope of a theoretical one-hit dilution curve. *B*, *C*, and *D* are experimental curves of tobacco streak virus infectivity from lesion counts on, respectively, *Dolichos biflorus*, *Vigna cylindrica*, and *D. lablab*. *E* is the infectivity dilution curve of Tulare apple mosaic virus from lesion counts on *Phaseolus vulgaris*. Lesion numbers are averages of total lesions on 3 replicate half-leaves from each of 3–5 trials on each host.

separate portions of a purified preparation were resuspended in different buffers to test their effect on stability *in vitro*. All buffers were at pH 8.0 and the virus was stored at 2°. In 148 days, less infectivity had been lost in borate buffer than in phosphate or Tris (Table 2). Borate buffer therefore was used to resuspend the virus pellets from the final two high-speed centrifugation steps for most of the work except serology.

Loss of infectivity on storage was accompanied by the appearance of small amounts of flocculant precipitate. When this precipitate was removed, there was no change in the absorption spectrum of the remaining virus, although the total absorbance was lower. The specific infectivity was somewhat less than that of fresh preparations, although variations in susceptibility of the assay plants may also have been involved.

TABLE 2

THE EFFECT OF DIFFERENT BUFFERS (0.02 *M*, pH 8.0) ON THE RETENTION OF INFECTIVITY OF PURIFIED TSV AT 2°[a]

Buffer	Lesions per half-leaf produced by preparations stored the following number of days		
	0	110	148
Phosphate	79	19	6
Borate	68	56	19
Tris	85	3	0.3

[a] Each contained ca. 875 μg of virus per milliliter; each was diluted 1:1000 in phosphate buffer for inoculation to *Dolichos biflorus*.

Electron Microscopy

Purified TSV, when mixed with potassium phosphotungstate (PTA), sprayed onto carbon-coated Formvar films, and examined with a JEM-7 electron microscope, was markedly disrupted. The virus in small drops of mixture placed on films without spraying was less distorted, but the PTA disrupted the virus rather quickly. Formalin (1%) was tried as a fixative, but preparations were little better than unfixed material. Fixation of the virus in 1% glutaraldehyde for 5–10 minutes before mixing with 2% PTA, however, resulted in particles much less flattened and distorted (Fig. 2) than in unfixed preparations. A small amount of purified tobacco mosaic virus was added to some preparations as a size standard.

Purified preparations consisted of uniform particles with a diameter of about 28 mμ. A fairly high proportion of the particles were not quite isodiametric, probably because

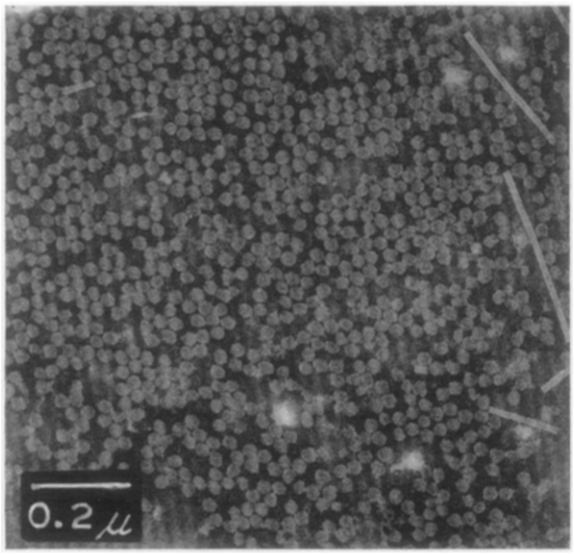


FIG. 2. Tobacco streak virus negatively stained with PTA. The tobacco mosaic virus was added as a size standard.

some uneven flattening occurred during drying.

Density Gradient Centrifugation

To detect inhomogeneity in purified preparations of TSV, appropriate dilutions were layered on gradients of 10–40% sucrose in 0.02 *M* borate buffer, pH 8.0. Gradients were prepared with a modification of an apparatus described by Stace-Smith (1965). After centrifugation in a Spinco SW 25.1 rotor, contents of the tubes were scanned with an ISCO density gradient fractionator (Instrumentation Specialties Co., Lincoln, Nebraska) with a recording UV (254 mμ) densitometer.

Preparations centrifuged 3 hours or longer at 25,000 rpm in 0.02 *M* borate buffer, pH 8.0, showed two closely spaced, but distinct, zones. When purified preparations of TSV were run in a Spinco model E analytical centrifuge, the two components had sedimentation coefficients of about 72 S and 96 S. Because they were closely spaced, samples were taken from the upper half of the upper zone and the lower half of the lower zone in density gradient tubes. Only samples from the lower zone were infectious.

The UV absorption curves were similar for each zone. The $A_{260:280}$ ratio for the upper zone was 1.49 ± 0.06; the ratio for the lower zone was 1.53 ± 0.06.

Approximately the same scanning pattern was found for several different preparations of the HF strain of TSV, with the upper (noninfectious) zone in lower concentration than the lower zone. Other strains were purified and subjected to density gradient centrifugation to see whether the same pattern was characteristic of all strains of the virus.

The proportions of the two components differed among the strains (Fig. 3). For each strain at least one other scanning pattern was obtained on a separate preparation of virus, and only minor differences were found in patterns for any one strain. Preparations of most strains contained less upper component than strain HF. Strain M, however, was unusual in producing more of the noninfectious upper component than of the lower component.

Strain M was used to investigate some factors that might influence the proportion of the two components. The temperature of incubation of inoculated plants or the host in which the virus was produced did not greatly affect the proportion of the two components (Fig. 4). Upper, symptomless

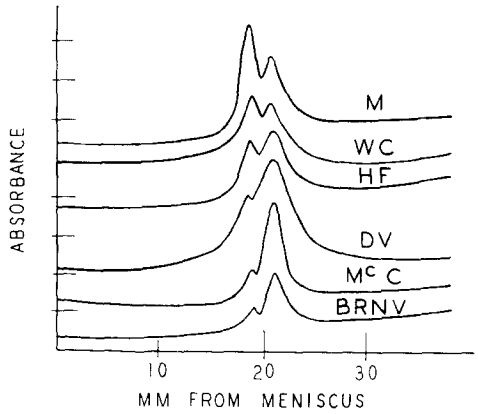


FIG. 3. Superimposed UV-scanning patterns of sucrose density gradient tubes on which purified preparations of six strains of tobacco streak virus had been layered. Tubes were centrifuged 4 hours at 25,000 rpm. Scale on the left indicates 0.1 absorbance units.

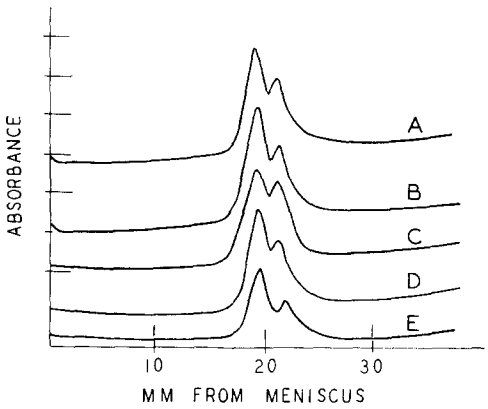


FIG. 4. Superimposed UV-scanning patterns of sucrose density gradient tubes on which TSV strain M had been layered before centrifugation for 4 hours. Curve *A*, purified from plants grown at 20°; *B*, purified from plants grown at 28°; *C*, purified from "recovered" leaves of plants infected 6 weeks; *D*, a purified preparation after storage at 2° for 6 months; *E*, purified from cucumber rather than tobacco. The scale on the left indicates 0.1 absorbance units.

leaves of plants infected for about 6 weeks yielded a preparation with only slightly less of the noninfectious component than a preparation from plants incubated for only a few days (Fig. 4) after inoculation. Storage of purified virus for several months at 2° did not alter the proportions of the two components, although as much as one-third of the total material was removed as an insoluble, denatured precipitate.

Biological Properties of Upper and Lower Components

Wood and Bancroft (1965) described an activating property of noninfectious particles of bean pod mottle and cowpea yellow mosaic viruses. This and the multiple-hit type of dilution curve found for TSV suggested testing the effect of the noninfectious component on the infectivity of the infectious component.

Trials were run with, as the infectious component, strain HF virus that had not been subjected to density gradient centrifugation and thus consisted of a natural mixture of the two types of particles. Upper component of the M strain, freed from sucrose by high speed centrifugation, was added to some inocula. Amounts of virus were standardized on the basis of absorbance at 260 mμ. Averages of three representative trials with purified virus are given in Table 3. The upper component markedly stimulated infection of *Vigna cylindrica* by TSV; the effect was less pronounced on *Dolichos biflorus*.

TABLE 3

THE EFFECT OF TOP COMPONENT ON THE INFECTIVITY OF UNFRACTIONATED TOBACCO STREAK VIRUS FOR TWO HOSTS

Inoculum	Average number of lesions per half-leaf	
	Vigna cylindrica	*Dolichos biflorus*
Unfractionated TSV (A_{260} = 0.004)	34	34
Unfractionated TSV (A_{260} = 0.004) plus upper component (A_{260} = 0.02)	90	45
Upper component (A_{260} = 0.02)	0	0

Purification of Tulare Apple Mosaic Virus

The similarity of symptoms and properties of Tulare apple mosaic virus (TAMV) to TSV suggested that the same purification scheme might be effective. Mink *et al.* (1963) reported that TAMV purified by *n*-butanol clarification and centrifugation retained infectivity in only a narrow pH range centering around pH 5.0. Even so, purified preparations had a half-life at 22° of only 13 hours.

In my work, infectivity of TAMV in crude tobacco leaf extracts was stabilized well by 0.02 *M* 2-ME, but extracts containing only 0.02 *M* DIECA became noninfectious in 4 hours. Infected leaves were homogenized in 0.02 *M*, pH 8.0 phosphate buffer containing 0.02 *M* DIECA in addition to 0.02 *M* 2-ME to avoid brown-colored virus pellets. The same schedule for purification was used as for TSV.

Infectivity of purified TAMV was about equal to that of TSV; preparations diluted to A_{260} of 0.0004–0.002 gave an average of two to ten lesions per half-leaf of Bountiful bean. The infectivity dilution curve of TAMV resembled that for TSV in being abnormally steep. There was a four- or fivefold decrease in infectivity with a twofold dilution (Fig. 1, curve *E*).

The UV absorption curve of purified TAMV was typical for nucleoprotein. The $A_{260:280}$ ratio, however, averaged 1.36 for five preparations (1.31–1.39), which was lower than that of any preparation of TSV.

There was no apparent loss of infectivity in several preparations of TAMV in 0.02 *M* phosphate buffer, pH 8.0, held for 10–12 days at 2°. There was partial loss of infectivity after storage for several months at 2°, but at 25°, in phosphate or borate buffer, the virus lost most of its infectivity in 3 or 4 days. In Tris buffer, however, the virus was appreciably more stable, losing less than one-half its infectivity in 5 days at 24°.

After 4 hours of density gradient centrifugation there were two closely spaced zones with sedimentation coefficients of about 85 S

and 95 S with virus in 0.02 *M* Tris buffer, pH 8.0 (Fig. 5).

Purified preparations for electron microscopy were fixed with 1% glutaraldehyde for 5–10 minutes and then mixed with an equal volume of 2% PTA. Drops applied to carbon-coated Formvar films were drawn off within a few seconds. Negatively stained TAMV was spherical and somewhat larger than TSV. Most particles were 32–34 mμ in diameter. About 0.35% of the particles were about twice as long as wide (Fig. 6).

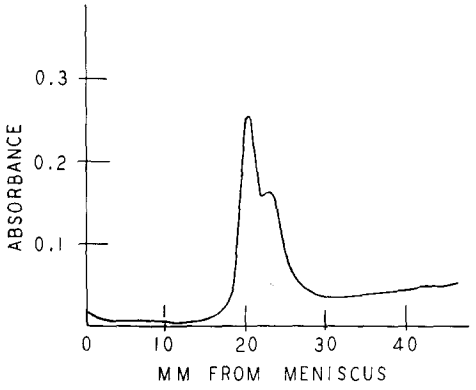


FIG. 5. Ultraviolet (254 mμ) scanning pattern of purified Tulare apple mosaic virus centrifuged 4 hours at 25,000 rpm in a sucrose density gradient tube.

DISCUSSION

With both TSV and TAMV the difficulty in preparing undisrupted virus for electron microscopy suggested that the protein coat of the virus was fragile. Possibly the instability of the viruses in tissue extracts is related to coat fragility. Both viruses are also relatively unstable when purified. Instability, therefore, is a property that cannot be explained solely on the basis of sensitivity to oxidized phenolic compounds.

Mink *et al.* (1966) reported that the

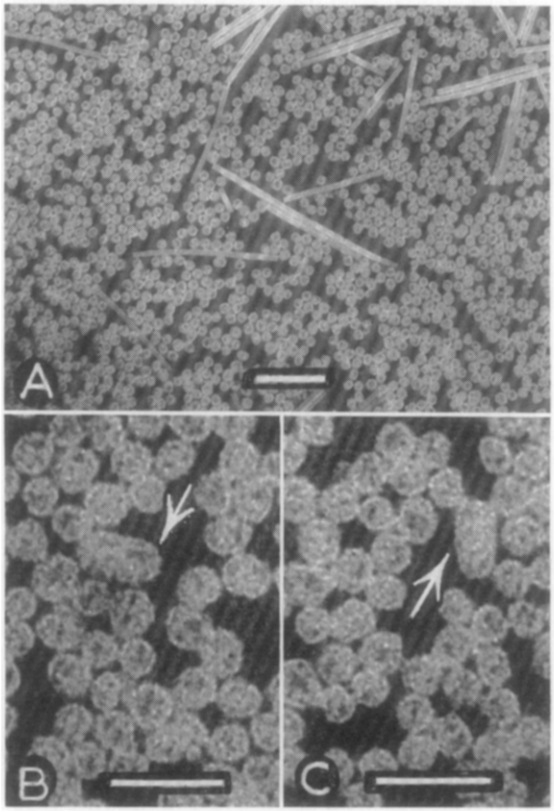


FIG. 6. Electron micrographs of Tulare apple mosaic virus, fixed in 1% glutaraldehyde and stained with PTA. The TMV was added as a size standard in (A). (B) and (C) show elongated particles (arrows) found occasionally with the normal isometric particles. The scale in (A) represents 0.2 μ; in (B) and (C), 0.1 μ.

number of zones obtained when the bean red node strain of TSV was subjected to density gradient centrifugation depended on the length of time plants had been infected. Preparations from plants infected 4 weeks separated into four zones. Infectivity was associated with the two more rapidly sedimenting zones, with sedimentation coefficients of 103 S and 123 S. These figures are appreciably higher than those found for TSV in this work, and only two components were found in preparatons of each of six strains of TSV. Neither the host species, tobacco or cucumber, or the length of time tobacco was infected significantly affected the proportions of each component. None of the preparations were made from bean, however, so it is not known whether some of the components found by Mink *et al.* (1966) do not occur in tobacco or cucumber, or whether they were selectively removed during purification.

There are a number of examples, as discussed by Schneider and Diener (1966), of viruses that are accompanied by similar particles that sediment more slowly in density gradient centrifugation and that are not infectious. These contain less nucleic acid than the infectious particles. Judging from the ratios of UV absorbance at 260 and 280 mμ, the two types of TSV particles contain nearly the same amounts of nucleic acid. The various components of several other viruses have been reported to contain appreciably different amounts of nucleic acid and sediment at quite different rates. Particles of squash mosaic virus, for example, had sedimentation coefficients of 57 S, 95 S, and 118 S (Mazzone *et al.*, 1962); those of wild cucumber mosaic, 53 S, 62.5 S, and 119 S (Yamazaki and Kaesberg, 1960); those of tobacco ringspot virus 53 S, 94 S, and 118 S (Stace-Smith *et al.*, 1965).

A peculiar property of the noninfectious, middle component particles of bean pod mottle and cowpea yellow mosaic viruses, described by Wood and Bancroft (1965), is that they stimulate infection by bottom component particles. This property is apparently possessed by TSV noninfectious particles also. Wood and Bancroft (1965) suggested that nucleic acid was involved in this activation because top component particles, lacking nucleic acid, were not effective. The biological function of such stimulatory, noninfective particles is not known, but it seems unlikely that production of such particles would persist as a strain characteristic if they were merely defects in assembly. Defects might also result in a greater variety of particle types than seem to occur with TSV, in which the upper component is similar in sedimentation rate for each of the strains investigated.

There may be a relation between a multiple-hit dilution curve and stimulation of infectivity by noninfective virus based on a requirement for more than one particle at an infection site. This seems to be conditioned by the host as well as by specific properties of the virus particles involved. The TSV infectivity dilution curve from lesion counts on *Dolichos biflorus* was a normal one-hit curve and there was only slight stimulation of infective virus by noninfective virus on this host. Lesion counts on *Vigna cylindrica*, on the other hand, resulted in a dilution infectivity curve that was abnormally steep. Noninfective virus markedly stimulated infection by infective virus on this host. This is evidence that factors other than, or in addition to, nucleic acid deficiency in the virus particles are involved in the stimulation.

Schneider and Diener (1966) thought that the middle component of tobacco ringspot virus was probably synthesized concurrently with bottom component. They found, also, more middle component in leaves infected 2½–3 days than in leaves infected 6–7 days. The data for TSV showed only minor differences in proportions of the two components resulting from variations in culture conditions or time. The periods were relatively long, however, and it is possible that the production of the two types of particles had reached an equilibrium within the first 4 or 5 days after infection.

The terms "top" and "bottom" component have been avoided because the term "top component" has acquired the connotation of referring to nucleic acid-free particles accompanying a number of other viruses. The upper zone particles of TSV resemble and behave as do "middle component" particles with other viruses. TSV and TAMV

are unusual in not producing a top component free of nucleic acid. It is possible, however, that a top component is produced but that it is too fragile to withstand the purification process.

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REFERENCES

COSTA, A. S., and CARVALHO, ANA M. B. (1961). Studies on Brazilian tobacco streak. *Phytopathol. Z.* **42,** 113–138.

DEVERGNE, J. C. (1961). Reconnaissance de plusieurs virus dans une population de *Trifolium pratense* L. naturellement infectée. *Ann. Épiphyties* **12,** 219–236.

DUNN, D. B., and HITCHBORN, J. H. (1965). The use of bentonite in the purification of plant viruses. *Virology* **25,** 171–192.

FULTON, R. W. (1949). Virus concentration in plants acquiring tolerance to tobacco streak. *Phytopathology* **39,** 231–243.

FULTON, R. W. (1959). Purification of sour cherry necrotic ringspot and prune dwarf viruses. *Virology* **9,** 522–535.

FULTON, R. W. (1962). The effect of dilution on necrotic ringspot virus infectivity and the enhancement of infectivity by noninfective virus. *Virology* **18,** 477–485.

GOLD, A. H. (1961). Antihost serum improves plant virus purification. *Phytopathology* **51,** 561–565.

MAZZONE, H. M., INCARDONA, N. L., and KAESBERG, P. (1962). Biochemical and biophysical properties of squash mosaic virus and related macromolecules. *Biochim. Biophys. Acta* **55,** 164–175.

MINK, G. I., BANCROFT, J. B., and NADAKAVUKAREN, M. J. (1963). Properties of Tulare apple mosaic virus. *Phytopathology* **53,** 973–978.

MINK, G. I., SAKSENA, K. N., and SILBERNAGEL, M. J. (1966). Purification of the bean red node strain of tobacco streak virus. *Phytopathology* **56,** 645–649.

PATINO, G., and ZAUMEYER, W. J. (1959). A new strain of tobacco-streak virus from peas. *Phytopathology* **49,** 43–48.

SCHNEIDER, I. R., and DIENER, T. O. (1966). The correlation between the proportions of virus-related products and the infectious component during the synthesis of tobacco ringspot virus. *Virology* **29,** 92–99.

SCOTT, H. A., VINCENT, M., and ZAUMEYER, W. J. (1961). Serological studies of red node and pod mottle viruses. *Phytopathology* **51,** 755–758.

STACE-SMITH, R. (1965). A simple apparatus for preparing sucrose density-gradient tubes. *Phytopathology* **55,** 1031.

STACE-SMITH, R., REICHMANN, M. E., and WRIGHT, N. S. (1965). Purification and properties of tobacco ringspot virus and two RNA-deficient components. *Virology* **25,** 487–494.

THOMAS, H. R., and ZAUMEYER, W. J. (1950). Red node, a virus disease of beans. *Phytopathology* **40,** 832–846.

WOOD, H. A., and BANCROFT, J. B. (1965). Activation of a plant virus by related incomplete nucleoprotein particles. *Virology* **27,** 94–102.

YAMAZAKI, H., and KAESBERG, P. (1960). Biophysical and biochemical properties of wild cucumber mosaic virus and of two related virus-like particles. *Biochim. Biophys. Acta* **51,** 9–18.